

Shahbano Farooq

+1-778-918-4655

shahbano_farooq@hotmail.com

www.linkedin.com/in/shahbanofarooq



EDUCATION

- **Phd Student** (Sept 2025 – To Date) Simon Fraser University – IAT, Vancouver, BC, Canada.
- **M.Sc Computer Science** (Jan 2012 – Apr 2014) from University of Calgary – Calgary, AB, Canada.
- **B.Sc (Honors) Computer Science** (Jan 1999 – Dec 2002) from Kinnaird College for Women – Lahore, Pakistan.

Awards

- **HEA Fellowship**: I received the Innovative Teaching and Learning Certification from AdvanceHE, London. It is in recognition of my innovative pedagogy, student-centered learning, and evidence-based teaching approaches as an Instructor at Zayed University.
- **Blackboard Academy Certified Associate**: I earned the certificate demonstrating proficiency in leveraging the Blackboard (Learning management system) tools to design, deliver, and manage technology-enhanced learning experiences that support student engagement and success.
- **Queen Elizabeth 2 Scholarship**: I was awarded this scholarship during my M.Sc. studies at the University of Calgary, Canada, in recognition of academic excellence and research achievement in designing a mind mapping exploratory visualization tool to support co-design.
- **K.K Aziz Gold Medal**: I received this gold medal for securing 1st position in my B.Sc. at Kinnaird College for Women, Lahore, Pakistan, in recognition of outstanding academic performance.

WORK EXPERIENCE

Teaching Experience at Zayed University - UAE (Fall 2017 – Spring 2025)

I have facilitated university level programming courses including; Introduction to IT (Fall 2017 till Fall 2020), Applied Algorithmic Thinking (Fall 2023 & Fall 2024), Programming Fundamentals (Spring 2021 till Spring 2023), Data Structures and Algorithms (Spring 2025), and Machine Learning (Fall 2024). I have heavily contributed to content creation and assessments to incorporate authentic learning experiences and experiential learning. In addition to teaching, I have supervised and mentored undergraduate internships (Spring 2022 & Fall 2022), providing technical guidance and feedback on work products. As a course coordinator for the Introduction to IT and Data Structures and Algorithms courses, I managed course planning, content, schedules, assessments, and communication between instructors.

Teaching Experience - Lab Instructor - University of Calgary (Winter 2012, Spring 2013)

I taught functional programming in Haskell to computer science students, highlighting its declarative paradigm, immutability, and use of higher-order functions, which contrasts with traditional object-oriented approaches. In parallel, I instructed object-oriented visual programming using Alice to general education students, focusing on objects, classes, and interactive design to develop foundational programming skills in a hands-on lab environment.

Technical Content Developer at Etisalat - UAE (Spring 2005- Spring 2007)

I was responsible for writing functional specifications and concept guides, as well as content creation and web development for internal newsletters and online help resources for the main telecommunication company in UAE. I also developed training materials and conducted training sessions for the new CBCM system for managing telecom services. One of my key contributions was designing an Appraisal Management System for the IT department in MS Access, using VBA to incorporate business knowledge. This application enabled management to set clear goals and objectives for the department, teams, and individuals, bridging the gap between targets and achievements and supporting effective performance evaluation and compensation. Throughout this work, I further developed skills in coordination, problem solving, and presentation.

Software Development and Support Engineer at Emircom – UAE (Fall 2003 – Fall 2004)

In this role, I was responsible for the full stack development of a help desk application system, while ensuring comprehensive documentation of all processes. The developed solution provided a web interface for managing customer service requests and complaints, workforce assignments, and escalation management. I built on a three-tier MVC architecture using J2EE components, the

front end was developed with JSP pages and JavaScript, while business logic was implemented in Java Beans and Servlets, and data was stored in MS SQL Server. Additionally, I developed a Java Applet to generate sales reports and charts in Excel, enabling management to analyze monthly, quarterly, and yearly sales performance. Through this work, I strengthened skills in decision-making, project planning, data modeling, application deployment, and resolving production issues.

System Analyst \ Programmer at ESP GITS – Pakistan (Spring 2002 – Fall 2003)

As a beginner level developer, I was responsible for analysis, prototyping, application design, data modeling, programming, and maintaining comprehensive documentation as a part of team. In collaboration, we developed a web application to facilitate credit approvals for MCB (Muslim Commercial Bank) managers, automating the bank's credit approval process using J2EE components and an Oracle 9i database. Additionally, we developed a messenger application for delivering educational lectures to a school in Saudi Arabia. Built with Visual Basic .Net using object-oriented concepts and an MS SQL Server database via ADO.Net.

RESEARCH EXPERIENCE

I recently contributed to a book on Object-Oriented programming in Python, presenting design and development concepts such as use cases, class diagrams, data persistence, and GUI development for modular design. I scaffolded learning with simpler steps that developed towards higher order thinking skills, as sample code snippets build towards a real-world case study. Besides co-contributing to most chapters, my sole contribution is on the chapter on Introduction to Machine Learning, that shows how machine learning applies object-oriented structures to support modular design.

Mathew, S. S., Kuhail, M. A., Hadid, M., & Farooq, S. (2025). The object-oriented approach to problem solving and machine learning with Python. Taylor and Francis Group.

I coauthored a book chapter on chatbot usability which identifies key usability factors applied in chatbot design, various application domains where chatbots have been utilized, and future directions for chatbot design. The most important design considerations identified are; conversational flow, personalization, context awareness, transparency, accessibility, and privacy to support a more human-like conversation.

Kuhail, M. A., Farooq, S., & Almutairi, S. (2023). Recent Developments in Chatbot Usability and Design Methodologies. Trends, Applications, and Challenges of Chatbot Technology, 1-23.

As a collaborator on the survey on immersive technologies in an educational setting, I investigated the pedagogical impact and design considerations of AR, VR, and mixed reality research. Important findings from this paper that support my current research are that spatial representation in learning improves retention and comprehension. In addition, embodiment such as gestures, navigation, and movement enhance engagement. This work strengthened my expertise in evaluating interaction modalities and their role in shaping user experience, directly supporting my PhD focus on Human–Computer Interaction

Kuhail, M.A.; ElSayary, A.; Farooq, S.; & Alghamdi, A. (2022). Exploring Immersive Learning Experiences: A Survey. Informatics 2022, 9, 75.

I also co-authored a literature review on visual programming interfaces to investigate how visual interfaces support application development for non-programmers. The focus of this research was to classify types of interfaces, interaction styles, and application of usability principles in design and evaluation. With the help of this paper, we provided a taxonomy for classification of visual programming interfaces for non-programmers.

Kuhail, M. A., Farooq, S., Hammad, R., & Bahja, M. (2021). Characterizing Visual Programming Approaches for End-User Developers: A Systematic Review. IEEE ACCESS, VOL 9, 14181-14202.

During my M.Sc, my research focus was on exploring collaborative design methods for visualization creation. I investigated how pairing domain experts with visualization designers can support the development of effective visualizations by facilitating shared understanding, knowledge exchange, and iterative design. My work examined strategies for structuring these collaborative sessions, the roles of participants, and how the interaction can support the design process. Therefore, I designed a tool, PairedVis that had three significant views, the data view, the mind mapping view, and the visualization design view. Mind mapping the relationships between the data columns encouraged contributions from the domain expert in sharing his knowledge and expertise with the visualization designer. Whereas, mapping visualisation templates and corresponding visual variables to data columns encouraged contributions from the visualisation designer. The research contributed insights into co-design practices and emphasized the importance of communication and negotiation between domain expertise and visualization expertise in producing meaningful and interpretable visual representations of data.

Farooq, S., Carpendale, S., and Maurer, F. (2016). Pairing for Designing Visualizations. In Surfnet: Designing Digital Surface Applications. (Maurer, F., Ed.). University of Calgary, Calgary, Canada, SurfNet, pages 155-168, March. ISBN 978-0-88953-388-2-(PDF).